

Riemannian stochastic optimization for sufficient dimension reduction

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Abstract

Gradient-based sufficient dimension reduction methods face a fundamental tradeoff: computing local gradients in the ambient space yields closed-form solutions but suffers from the curse of dimensionality, while iterative refinement in the projected space improves statistical efficiency at $O(n^2p)$ cost per iteration. We show that minimizers of the Minimum Average Variance Estimation (MAVE) criterion recover the span of the regression function’s gradients—the same target as the Outer Product of Gradients method—but through local regression in the projected space. We then reformulate MAVE as a Riemannian maximization problem on the Stiefel manifold and derive a closed-form gradient, enabling efficient stochastic optimization. The resulting algorithm, SMAVE, combines mini-batch Riemannian gradient ascent with adaptive k -nearest neighbor localization that evolves with the subspace estimate. On synthetic benchmarks, SMAVE matches or exceeds the accuracy of existing methods while running 10–50× faster; on real regression tasks, these gains translate to improved prediction with speedups exceeding three orders of magnitude.

1. Introduction

Sufficient dimension reduction (SDR) seeks a low-dimensional linear projection $\mathbf{B}^\top X$ of high-dimensional covariates $X \in \mathbb{R}^p$ that preserves all information relevant for predicting a response Y . Unlike unsupervised techniques such as principal component analysis, SDR exploits the supervised structure: the projection captures label-relevant variation rather than overall data variance. When such a projection exists with $d \ll p$, the intractable problem of estimating the regression function $\mathbb{E}[Y | X]$ in $\mathbb{R}^p$ reduces to the tractable problem of estimating $\mathbb{E}[Y | \mathbf{B}^\top X]$ in $\mathbb{R}^d$.

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Beyond computational savings, this projection provides an interpretable, low-dimensional representation of the covariates for the supervised learning problem.

This goal places SDR within a broader family of supervised representation learning methods. Metric learning provides a concrete example: learning a Mahalanobis distance $d_M(x, x') = \|x - x'\|_M$ adapted to prediction tasks (Ghosh et al., 2022; Kulis et al., 2013) reduces to learning a linear projection when the metric matrix M has low rank. If $\text{rank}(M) = d \ll p$, then $M = \mathbf{B}\mathbf{B}^\top$ for some $\mathbf{B} \in \mathbb{R}^{p \times d}$, and $d_M(x, x') = \|\mathbf{B}^\top(x - x')\|_2$ —precisely the Euclidean distance in the projected space that SDR methods seek to identify.

Two broad families of methods have emerged for estimating the subspace spanned by $\mathbf{B}$ from n independent samples, each navigating a fundamental tension between statistical flexibility and computational cost. Inverse regression approaches—including Sliced Inverse Regression (SIR; Li, 1991), Sliced Average Variance Estimation (SAVE; Cook & Weisberg, 1991), and Directional Regression (DR; Li & Wang, 2007)—exploit moment conditions on $\mathbb{E}[X | Y]$ to achieve $\sqrt{n}$ -consistency with low computational cost, but require linearity or constant variance conditions that often fail in practice. Gradient-based methods offer greater flexibility by estimating the subspace directly from local derivative information, avoiding these distributional assumptions. The Average Derivative Estimator (Härdle & Stoker, 1989) and Outer Product of Gradients (OPG; Xia et al., 2002) construct subspace estimates from pointwise gradient estimates, while Minimum Average Variance Estimation (MAVE; Xia et al., 2002) formulates the problem as local polynomial regression. However, this flexibility comes at a computational price: the kernel bandwidth scale causes neighborhoods to become exponentially sparse when p is large. Refined variants—RMAVE (Xia et al., 2002) and structure-adaptive OPG (Hristache et al., 2001; Dalalyan et al., 2008)—address this by localizing in the projected space $\mathbb{R}^d$, but still require $O(n^2p)$ operations due to dense pairwise kernel weights. A separate line of work, initiated by Fukumizu et al. (2009a), characterizes dimension reduction subspaces through conditional independence in reproducing kernel Hilbert spaces (Wu et al., 2019; Park et al., 2023), offering a distinct perspective that we do not pursue here.

Since the target is a subspace rather than a specific matrix, the natural parameter space is the Grassmann manifold $\text{Gr}(p, d)$ of d -dimensional linear subspaces of $\mathbb{R}^p$. For computation, we represent each subspace by a matrix with orthonormal columns, optimizing over the Stiefel manifold $\text{St}(p, d) = \{\mathbf{B} \in \mathbb{R}^{p \times d} : \mathbf{B}^\top \mathbf{B} = \mathbf{I}_d\}$. Riemannian optimization on these matrix manifolds is well-established in numerical linear algebra (Edelman et al., 1998; Absil et al., 2008) and increasingly adopted in machine learning, from decentralized training with orthogonality constraints (Chen et al., 2021) to parameter-efficient fine-tuning of large language models (Park et al., 2025). We bring these tools to sufficient dimension reduction, demonstrating that scalable Riemannian stochastic gradient methods can unlock both statistical and computational gains for foundational problems in nonparametric statistics.

We propose **SMAVE** (Stochastic MAVE), a scalable algorithm that combines Riemannian optimization with adaptive localization in the projected space to achieve both statistical efficiency and computational tractability.

Contributions.

1. **Gradient alignment.** We show that MAVE minimizers approximate the subspace spanned by the gradients of the regression function—the same target as OPG, but recovered through local regression rather than gradient aggregation. This justifies the MAVE criterion from a gradient-based perspective (Propositions 2.4–2.5).
2. **Riemannian reformulation.** We reformulate the MAVE criterion as maximizing a smooth objective on $\text{St}(p, d)$ (Proposition 2.7) and derive its Riemannian gradient in closed form, with no projection step required (Proposition 4.1). The objective is $\mathcal{O}(d)$ -invariant, confirming it descends to the Grassmannian.
3. **Scalable algorithm.** Building on this reformulation, SMAVE combines mini-batch Riemannian SGD with sparse k -nearest neighbor (k -NN) weights in the projected space, periodically refreshed as the subspace estimate evolves. This reduces per-epoch complexity from $\mathcal{O}(n^2 p)$ to $\mathcal{O}(n \log n)$.
4. **Empirical validation.** SMAVE matches or exceeds RMAVE accuracy while running 10–50 $\times$ faster, with particularly strong gains when p is large. These improvements translate to better prediction accuracy on real regression tasks.

Organization. Section 2 formalizes the SDR problem and introduces our local approximation framework. Section 3 reviews related work, including gradient-based SDR methods and Riemannian optimization. Section 4 presents the

SMAVE algorithm. Section 5 reports experiments on synthetic and real data. Proofs and additional results appear in the appendix.

2. Problem setup

2.1. Sufficient Dimension Reduction

Let (Y, X) be a random vector with $Y \in \mathbb{R}$ and $X \in \mathbb{R}^p$, and let $g(x) := \mathbb{E}[Y | X = x]$ denote the regression function. When p is large, estimating g directly suffers from the curse of dimensionality. The goal of sufficient dimension reduction is to find a matrix $\mathbf{B} \in \mathbb{R}^{p \times d}$ with $d \ll p$ such that

$$g(X) = \mathbb{E}[Y | \mathbf{B}^\top X]. \quad (1)$$

When such a projection exists, the intractable problem of estimating g in $\mathbb{R}^p$ reduces to estimation in the low-dimensional space $\mathbb{R}^d$.

Remark 2.1 (Central mean subspace). Condition (1) defines the *central mean subspace*, weaker than the classical *central subspace* requiring $Y \perp X | \mathbf{B}^\top X$. The central mean subspace suffices for regression and is the natural target of gradient-based methods (Xia et al., 2002).

The condition (1) exhibits a key invariance: it holds for $\mathbf{B}$ if and only if it holds for $\mathbf{B}\mathbf{A}$, for any invertible $\mathbf{A} \in \text{GL}_d(\mathbb{R})$, since $\mathbf{B}^\top X$ and $\mathbf{A}^\top \mathbf{B}^\top X$ generate the same σ -algebra. The natural parameter space is therefore the Grassmann manifold

$$\text{Gr}(p, d) = \{\text{span}(\mathbf{B}) : \mathbf{B} \in \mathbb{R}^{p \times d}, \text{rank}(\mathbf{B}) = d\},$$

consisting of all d -dimensional linear subspaces of $\mathbb{R}^p$.

For computation, we represent each subspace by a matrix with orthonormal columns. The Stiefel manifold $\text{St}(p, d)$ comprises all $p \times d$ matrices satisfying $\mathbf{B}^\top \mathbf{B} = \mathbf{I}_d$. This representation is not unique: $\mathbf{B}, \mathbf{B}' \in \text{St}(p, d)$ span the same subspace if and only if $\mathbf{B}' = \mathbf{B}\mathbf{Q}$ for some $\mathbf{Q} \in \mathcal{O}(d)$, yielding the quotient identification $\text{Gr}(p, d) \cong \text{St}(p, d)/\mathcal{O}(d)$. Optimizing over $\text{St}(p, d)$ simplifies implementation, provided the objective is $\mathcal{O}(d)$ -invariant—a property we verify below. The Riemannian optimization machinery (tangent spaces, gradients, and retractions on the Stiefel manifold) is detailed in Appendix A.

The natural approach to finding $\mathbf{B}$ is to minimize the prediction error on the Stiefel manifold:

$$\min_{\mathbf{B} \in \text{St}(p, d)} \{D(\mathbf{B}) := \mathbb{E}[(Y - \mathbb{E}[Y | \mathbf{B}^\top X])^2]\}.$$

Since $D(\mathbf{B}) = \sigma^2 + \mathbb{E}[(g(X) - \mathbb{E}[Y | \mathbf{B}^\top X])^2]$, where $\sigma^2 = \mathbb{E}[\text{Var}(Y | X)]$, minimizing $D(\mathbf{B})$ amounts to approximating $g(X)$ by the projected regression function $\mathbb{E}[Y | \mathbf{B}^\top X]$ in $L_2(\mathbb{P})$.

2.2. Local approximation analysis

Direct optimization is intractable since $\mathbf{B}$ appears inside the conditioning event. The MAVE approach (Xia et al., 2002) introduces a local surrogate: for $x \in \mathbb{R}^p$ and $u > 0$, define

$$D_{(x,u)}(\mathbf{B}) := \min_{a \in \mathbb{R}, \mathbf{b} \in \mathbb{R}^d} \mathbb{E}_{(x,u)} [(Y - a - \mathbf{b}^\top \mathbf{B}^\top (X - x))^2], \quad (2)$$

where $\mathbb{E}_{(x,u)}[\cdot]$ denotes expectation conditional on $X \in B(x, u) = \{y \in \mathbb{R}^p : \|y - x\| \leq u\}$. When u is small, the optimal a approximates $g(x)$ and $\mathbf{B}\mathbf{b}$ approximates the projection of $\nabla g(x)$ onto $\text{span}(\mathbf{B})$.

Let $P_{\mathbf{B}} = \mathbf{B}\mathbf{B}^\top$ and $G_{(x,u)} = \text{Cov}_{(x,u)}(X)$ denote the orthogonal projection and local covariance, respectively.

Assumption 2.2 (Smoothness). There exists $L > 0, u_0 > 0$ such that, for $\mathbb{P}$ -almost every x and all $u \in (0, u_0]$,

$$\mathbb{E}_{(x,u)} [|g(X) - g(x) - \nabla g(x)^\top (X - x)|^2] \leq \frac{L^2}{4} u^4. \quad (3)$$

Assumption 2.3 (Local covariance lower bound). There exists $\lambda \in (0, 1], u_0 > 0$ such that, for $\mathbb{P}$ -almost every x and all $u \in (0, u_0]$,

$$G_{(x,u)} \succeq \lambda u^2 I_p. \quad (4)$$

Assumption 2.2 holds when g has bounded second derivatives; Assumption 2.3 holds when X has a density bounded away from zero. The constraint $\lambda \leq 1$ is necessary (Appendix B).

Proposition 2.4 (Local scale separation). *Under Assumptions 2.2–2.3, for all $u \in (0, u_0]$:*

(i) *If $u \leq \frac{\lambda}{2L} \|(I_p - P_{\mathbf{B}})\nabla g(x)\|_2$, then*

$$D_{(x,u)}(\mathbf{B}) \geq \mathbb{E}_{(x,u)}[\sigma^2(X)] + \frac{\lambda u^2}{2} \|(I_p - P_{\mathbf{B}})\nabla g(x)\|_2^2.$$

(ii) *If $\nabla g(x) \in \text{span}(\mathbf{B})$, then*

$$D_{(x,u)}(\mathbf{B}) \leq \mathbb{E}_{(x,u)}[\sigma^2(X)] + \frac{L^2 u^4}{4}.$$

The proposition establishes a *scale separation*: aligned subspaces incur excess risk $O(u^4)$, while misaligned ones pay $\Omega(u^2)$. Integrating over x yields a global characterization. Define $D_u(\mathbf{B}) := \int D_{(x,u)}(\mathbf{B}) \mathbb{P}(dx)$ and the gradient outer product matrix $\mathcal{M} := \mathbb{E}[\nabla g(X) \nabla g(X)^\top]$.

Proposition 2.5 (Global gradient alignment). *Under Assumptions 2.2–2.3, any minimizer $\mathbf{B}$ of D_u satisfies*

$$E(\mathbf{B}, \mathcal{M}) := \text{tr}((I_p - P_{\mathbf{B}})\mathcal{M}) \leq \frac{9L^2}{2\lambda^2} u^2.$$

Geometric interpretation. The discrepancy $E(\mathbf{B}, \mathcal{M})$ measures the $\mathcal{M}$ -weighted projection error on the Grassmannian $\text{Gr}(d, p)$ of d -dimensional subspaces of $\mathbb{R}^p$. Writing $\mathcal{M} = \sum_i \mu_i v_i v_i^\top$ in its eigendecomposition, we have $E(\mathbf{B}, \mathcal{M}) = \sum_i \mu_i \sin^2 \theta_i$, where θ_i is the angle between v_i and $\text{span}(\mathbf{B})$. Minimizing $E(\mathbf{B}, \mathcal{M})$ thus seeks subspaces aligned with the dominant eigenvectors of $\mathcal{M}$. The matrix $\mathcal{M}$ is precisely the population objective of OPG; Proposition 2.5 shows that MAVE targets the same subspace—the top eigenspace of $\mathcal{M}$ —through local regression rather than gradient aggregation.

Remark 2.6. The localization can be relaxed from balls to cylinders $B_{\mathbf{A}}(x, u) = \{y : \|\mathbf{A}^\top (y - x)\| \leq u\}$ for any $\mathbf{A}$ with $\text{span}(\mathbf{B}) \subseteq \text{span}(\mathbf{A})$; localizing in the reduced space ($\mathbf{A} = \mathbf{B}$) avoids the curse of dimensionality, motivating the adaptive scheme in Section 4.

2.3. Empirical optimization problem

Let $(X_i, Y_i)_{i=1, \dots, n} \subset \mathbb{R}^p \times \mathbb{R}$ be i.i.d. random variables with common distribution $\mathbb{P}$. We estimate the local expectation $\mathbb{E}_{(x,u)}[\cdot]$ using the k -NN estimator: for a point x , we average over the set $\mathcal{N}_k(x)$ of the k nearest neighbors to x among $(X_1, \dots, X_n)$ with respect to the Euclidean norm.

Substituting into the integrated objective yields the empirical MAVE criterion: denote the local linear prediction residual by

$$r_{ij}(a, \mathbf{b}, \mathbf{B}) := Y_i - a - \mathbf{b}^\top \mathbf{B}^\top (X_i - X_j).$$

The empirical MAVE criterion is then

$$\min_{\mathbf{B} \in \text{St}(p, d)} \left\{ D_n(\mathbf{B}) = \sum_{j=1}^n \sum_{i=1}^n r_{ij}(\mathbf{B})^2 w_{ij} \right\}. \quad (5)$$

where $w_{ij} = k^{-1} \mathbf{1}_{i \in \mathcal{N}_k(X_j)}$ are weights selecting neighbors of X_j , and for each j , $(a_j, \mathbf{b}_j)$ are local linear regression coefficients:

$$(a_j, \mathbf{b}_j) = \arg \min_{a, \mathbf{b}} \sum_{i=1}^n r_{ij}(a, \mathbf{b}, \mathbf{B})^2 w_{ij},$$

Different weight schemes may be used (e.g., Gaussian kernel weights). Moreover, following the cylinder relaxation of Remark 2.6, we may use $\mathcal{N}_{k, \mathbf{A}}(X_j)$ —the k -NN under the semi-norm $\|x\|_{\mathbf{A}} = \|\mathbf{A}^\top x\|_2$ —for any $\mathbf{A}$ with $\text{span}(\mathbf{B}) \subseteq \text{span}(\mathbf{A})$. This leads to the adaptive procedure in Section 4.

We now reformulate (5) as a maximization problem amenable to Riemannian optimization. Define the local

moment vector and Gram matrix

$$\begin{aligned}\mu^{(j)} &= \sum_{i=1}^n w_{ij} \tilde{X}_i^{(j)} \tilde{Y}_i^{(j)} \in \mathbb{R}^p, \\ G^{(j)} &= \sum_{i=1}^n w_{ij} \tilde{X}_i^{(j)} (\tilde{X}_i^{(j)})^\top \in \mathbb{R}^{p \times p},\end{aligned}$$

where $\tilde{Y}_i^{(j)} = Y_i - \bar{Y}^{(j)}$, $\tilde{X}_i^{(j)} = X_i - \bar{X}^{(j)}$, and $\bar{Y}^{(j)}$, $\bar{X}^{(j)}$ are the weighted local means.

Proposition 2.7 (Stiefel reformulation). *Minimizing (5) over $\mathbf{B} \in \text{St}(p, d)$ is equivalent to maximizing*

$$F(\mathbf{B}) := \sum_{j=1}^n (\mu^{(j)})^\top \mathbf{B} (\mathbf{B}^\top G^{(j)} \mathbf{B})^{-1} \mathbf{B}^\top \mu^{(j)}. \quad (6)$$

Moreover, $F(\mathbf{B}\mathbf{Q}) = F(\mathbf{B})$ for any orthogonal $\mathbf{Q} \in \mathcal{O}(d)$, so F descends to a well-defined function on the Grassmannian $\text{Gr}(p, d)$.

3. Related work

We review gradient-based SDR methods and Riemannian optimization, situating our contributions within this literature.

MAVE. The original MAVE algorithm (Xia et al., 2002) minimizes the same objective D_n but with kernel weights $w_{ij} \propto K_h(\|X_i - X_j\|)$ rather than k -NN, and employs a different optimization strategy. Rather than updating $\mathbf{B}$ globally, MAVE uses coordinate descent, cycling through columns $\beta_1, \dots, \beta_d$. Each column update requires solving a constrained quadratic subproblem subject to orthogonality with the remaining columns, itself handled by alternating between local coefficient updates and a KKT system. In contrast, our Riemannian approach updates all columns simultaneously, maintaining orthogonality intrinsically via retraction rather than explicit constraints.

Extensions include dMAVE (Xia, 2007), which transforms Y through a density kernel to estimate the full central subspace rather than just the central mean subspace. Our framework could accommodate this extension, but we leave it for future work.

OPG. The Outer Product of Gradients method (Xia et al., 2002) estimates gradients via local linear regression in the ambient space. For each anchor X_j , OPG solves

$$\min_{a_j, b_j \in \mathbb{R}^p} \sum_{i=1}^n (Y_i - a_j - b_j^\top (X_i - X_j))^2 w_{ij},$$

with kernel weights $w_{ij} \propto K_h(\|X_i - X_j\|)$ computed in $\mathbb{R}^p$. The gradient estimate is $\hat{b}_j = (G^{(j)})^{-1} \mu^{(j)}$, and the

EDR subspace is recovered as the top d eigenvectors of

$$\hat{\Sigma}_{\text{OPG}} = \frac{1}{n} \sum_{j=1}^n \hat{b}_j \hat{b}_j^\top.$$

OPG admits a closed-form solution but computes neighborhoods in $\mathbb{R}^p$, suffering from the curse of dimensionality when p is large. When local covariances are isotropic ($G^{(j)} \propto I_p$), OPG and MAVE coincide (Proposition B.4). A k -NN variant of OPG is proposed in (Ausset et al., 2021), which reduces computational cost by restricting computations to selected neighbors.

RMAVE. Refined MAVE (Xia et al., 2002), as well as the structure-adaptive approach of Hristache et al. (2001); Dalalyan & Spokoiny (2008), addresses the dimensionality issue through iterative refinement. Given an estimate $\hat{\mathbf{B}}$, RMAVE recomputes kernel weights using proximity in the reduced space:

$$w_{ij} \propto K_h(\|\hat{\mathbf{B}}^\top (X_i - X_j)\|),$$

with $q = d$ in Silverman’s rule (Silverman, 2018), so that smaller bandwidths become feasible. The MAVE problem is then re-solved with these refined weights.

However, RMAVE inherits MAVE’s coordinate descent structure and adds significant overhead: each iteration requires $O(n^2 p)$ operations to recompute pairwise weights, and convergence typically demands tens to hundreds of outer iterations. Crucially, RMAVE relies on OPG for initialization; when p is large, this initialization is poor, limiting RMAVE’s effectiveness even with reduced-space refinement.

This $O(n^2 p)$ bottleneck is shared by all existing gradient-based SDR methods, including the structure-adaptive refinements above. Our approach combines RMAVE’s reduced-space localization with sparse k -NN weights and Riemannian optimization on $\text{St}(p, d)$, reducing complexity to $O(n \log n)$.

MADE. Minimum Average Deviance Estimation (Adraghi, 2018) extends MAVE to exponential family distributions by replacing the squared loss with local deviance, enabling application to binary and count responses. For continuous responses with Gaussian errors—the setting of all experiments in this paper—MADE with identity link reduces to standard MAVE, making RMAVE the appropriate iterative baseline. Beyond this statistical equivalence, MADE and SMAVE address orthogonal challenges: MADE generalizes the loss function while SMAVE addresses computational scalability. The MADE algorithm relies on Riemannian conjugate gradient with matrix exponentials for geodesic retraction, parallel transport of tangent vectors, and Newton–Raphson

iterations for local parameters—primitives that are orders of magnitude more expensive than our first-order stochastic updates with QR retraction. Including MADE would thus either duplicate the RMAVE comparison (Gaussian case) or disadvantage SMAVE on non-Gaussian data where squared loss is suboptimal. We note that the scalability techniques developed here could potentially be combined with MADE’s deviance-based objective in future work.

Kernel-based methods. Kernel Dimension Reduction (Fukumizu et al., 2009b) characterizes the central subspace via conditional covariance operators in reproducing kernel Hilbert spaces, avoiding explicit gradient estimation. While theoretically elegant, these methods involve different computational trade-offs (kernel matrix operations vs. local regression) and are not directly comparable to the MAVE family we focus on here.

Riemannian optimization. Riemannian optimization (Absil et al., 2008) extends gradient-based methods to smooth manifolds, requiring two ingredients: the Riemannian gradient and a retraction mapping tangent vectors back to the manifold. Convergence guarantees for Riemannian SGD are well established (Bonnabel, 2013; Zhang et al., 2016). Appendix A provides background on Riemannian geometry.

4. SMAVE algorithm

We now present SMAVE, which resolves the tension between statistical efficiency and computational scalability by: (i) replacing dense kernel weights with sparse k -NN weights in the projected space, and (ii) optimizing directly on $\text{St}(p, d)$ via Riemannian stochastic gradient ascent.

4.1. Adaptive sparse localization via k -NN

RMAVE computes weights $w_{ij} \propto K_h(\|\hat{\mathbf{B}}^\top(X_i - X_j)\|)$ for all n^2 pairs, though local regression requires only local information. We exploit this sparsity by defining

$$w_{ij} = \begin{cases} 1/k & \text{if } X_i \in \mathcal{N}_k(\hat{\mathbf{B}}, X_j), \\ 0 & \text{otherwise,} \end{cases} \quad (7)$$

where $\mathcal{N}_k(\mathbf{B}, X_j)$ denotes the k nearest neighbors of X_j under the projected distance $\|\hat{\mathbf{B}}^\top(\cdot - X_j)\|$. This reduces weight computation from $O(n^2)$ to $O(nk)$. Since neighborhoods depend on $\hat{\mathbf{B}}$, they become stale as the subspace estimate evolves. We rebuild the k -NN index every τ iterations via KD-tree on $\{\hat{\mathbf{B}}_t^\top X_i\}_{i=1}^n$, tree construction costs $O(dn \log n)$, which reduces to $O(n \log n)$ for fixed d . Between rebuilds, the fixed index introduces slight mismatch but avoids per-iteration overhead. Moderate values of τ balance gradient fidelity against computational cost (see Section 5).

Algorithm 1 SMAVE

Require: Data $\{(X_i, Y_i)\}_{i=1}^n$, dimension d , iterations T , refresh period τ

- 1: Initialize $\mathbf{B}_0 \in \text{St}(p, d)$; set $k \leftarrow \lfloor n^{4/(d+4)} \rfloor$ clipped to $[20, n/3]$
- 2: Build k -NN index on $\{\mathbf{B}_0^\top X_i\}$
- 3: **for** $t = 0, \dots, T - 1$ **do**
- 4: Sample mini-batch J_t ; compute $\mathbf{g}_t$ via (8); normalize $\mathbf{g}_t \leftarrow \mathbf{g}_t / \|\mathbf{g}_t\|_F$
- 5: Update $\mathbf{B}_{t+1}$ via (9)
- 6: **if** $t \bmod \tau = 0$ **then**
- 7: Rebuild k -NN index on $\{\mathbf{B}_{t+1}^\top X_i\}$
- 8: **end if**
- 9: **end for**
- 10: **return** $\mathbf{B}_T$

4.2. Riemannian stochastic gradient ascent

The following result, proved in Appendix B.5, provides the gradient required for optimization on $\text{St}(p, d)$.

Proposition 4.1 (Riemannian gradient). *Assume $\mathbf{B}^\top G^{(j)} \mathbf{B}$ is invertible for all j . The Riemannian gradient of F in (6) is*

$$\text{grad } F(\mathbf{B}) = 2 \sum_{j=1}^n \left(\mu^{(j)} - G^{(j)} \mathbf{B} u^{(j)} \right) (u^{(j)})^\top,$$

where $u^{(j)} = (\mathbf{B}^\top G^{(j)} \mathbf{B})^{-1} \mathbf{B}^\top \mu^{(j)}$.

At each iteration, we sample a mini-batch $J_t \subset \{1, \dots, n\}$ and form the stochastic gradient

$$\mathbf{g}_t = \frac{2n}{|J_t|} \sum_{j \in J_t} \left(\mu^{(j)} - G^{(j)} \mathbf{B}_t u^{(j)} \right) (u^{(j)})^\top, \quad (8)$$

which satisfies $\mathbb{E}[\mathbf{g}_t | \mathbf{B}_t] = \text{grad } F(\mathbf{B}_t)$. We normalize $\mathbf{g}_t \leftarrow \mathbf{g}_t / \|\mathbf{g}_t\|_F$ to stabilize step sizes, incorporate heavy-ball momentum, and update via QR retraction:

$$\mathbf{v}_{t+1} = \beta \mathbf{v}_t + \mathbf{g}_t, \quad \mathbf{B}_{t+1} = R_{\mathbf{B}_t}^{\text{QR}}(\alpha_t \mathbf{v}_{t+1}), \quad (9)$$

where $\alpha_t = \alpha_0 / (1 + \gamma t)$ is a decaying step size. This approach parallelizes naturally across batch elements.

Algorithm 1 summarizes the complete procedure. Implementation details, including the choice of k , are in Appendix C.

Remark 4.2 (Momentum implementation). The momentum vector $\mathbf{v}_t$ accumulates in ambient space without parallel transport between tangent spaces. This common simplification avoids $O(pd^2)$ transport costs per iteration; the QR retraction implicitly projects updates onto the manifold (Bécigneul & Ganeva, 2018).

5. Numerical experiments

5.1. Baselines and complexity

We compare four SDR methods enabling controlled ablations of our two contributions: sparse k -NN localization and Riemannian stochastic optimization. **OPG** and **RMAVE** are described in Section 3. **OPG-kNN** replaces OPG’s kernel weights with uniform k -NN weights in $\mathbb{R}^p$, isolating sparse localization. **SMAVE** combines both contributions: k -NN weights in the projected space $\mathbb{R}^d$ and Riemannian stochastic gradient ascent. We omit structure-adaptive OPG (Hristache et al., 2001; Dalalyan et al., 2008), which improves statistical efficiency but shares RMAVE’s $O(n^2p)$ cost.

Neighborhood sizes are calibrated to match effective kernel neighborhoods. OPG and RMAVE use Gaussian kernels with Silverman’s rule-of-thumb bandwidth $h \propto n^{-1/(q+4)}$ (Silverman, 2018), where q is the operating dimension ($q = p$ for OPG, $q = d$ for RMAVE). For k -NN methods, we set $k = O(n^{4/(q+4)})$ to match the expected number of points within the kernel’s effective support ($q = p$ for OPG-kNN, $q = d$ for SMAVE). Full implementation details are in Appendix C. Table 1 summarizes computational costs.

Table 1. Computational complexity. T : iterations; m : mini-batch size; k : neighbors; τ : k -NN refresh period.

Method	Per-iteration	Total
OPG	—	$O(n^2p)$
OPG-kNN	—	$O(pn \log n + nkp^2 + np^3)$
RMAVE	$O(n^2p)$	$O(Tn^2p)$
SMAVE	$O(mkpd + md^3)$	$O(Tmkpd + \frac{T}{\tau}dn \log n)$

OPG and RMAVE share an $O(n^2)$ bottleneck from dense weights; OPG-kNN eliminates this but retains $O(np^3)$ from p -dimensional solves. SMAVE avoids both: mini-batching reduces n^2 to mk , and operating in $\mathbb{R}^d$ replaces p^3 with d^3 , yielding near-constant runtime across (n, p) configurations.

5.2. Synthetic experiments

We consider the SDR model $Y = g(\mathbf{B}_0^\top \mathbf{X}) + \varepsilon$, where $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \Sigma)$, $\mathbf{B}_0 \in \text{St}(p, d)$ is a sparse EDR matrix (20% nonzero entries, QR-orthonormalized) with $d = 2$, and $\varepsilon \sim \mathcal{N}(0, 0.5^2)$.

Sparse EDR matrices arise naturally in high-dimensional settings where only a subset of covariates contributes to the response, aligning with the sufficient variable selection literature (Li, 2007). All methods compared are agnostic to the sparsity structure and do not exploit it explicitly.

Evaluation metric. Given the true EDR matrix $\mathbf{B}_0$ and an estimate $\hat{\mathbf{B}}$, we measure accuracy via the squared subspace

Table 2. Experimental design: 10 scenarios from 5 link functions $\times$ 2 covariance structures, with $n \in \{1000, 2000, 5000\}$ and $p \in \{10, 20, 50, 100\}$.

Factor	Levels
Link functions	Polynomial, Sinusoidal, Exponential, Interaction, Rational
Covariance Σ	Identity ($\mathbf{I}_p$); AR(1) ($\Sigma_{ij} = 0.5^{ i-j }$)
Sample size n	1000, 2000, 5000
Dimension p	10, 20, 50, 100
Replications	10 per configuration

distance

$$m^2(\hat{\mathbf{B}}, \mathbf{B}_0) = \|(I_p - \mathbf{B}_0 \mathbf{B}_0^\top) \hat{\mathbf{B}}\|_F^2, \quad (10)$$

which equals zero when the column spaces coincide and is bounded by d when they are orthogonal.

Table 3. Squared subspace distance m^2 (mean $\pm$ s.e. over 10 scenarios $\times$ 10 replications). **Bold**: best; underline: second best.

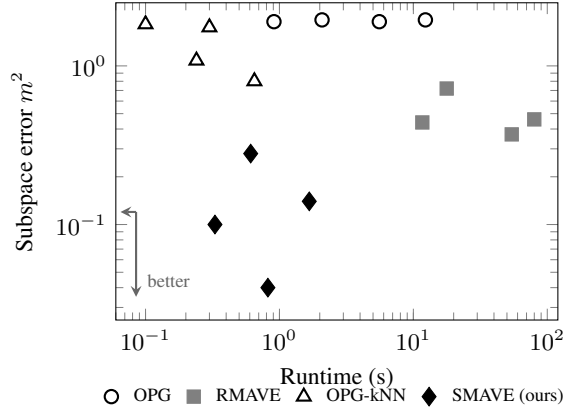
n	p	OPG	RMAVE	OPG-kNN	SMAVE
1000	10	0.08 $\pm$.01	0.01$\pm$.00	0.04 $\pm$.00	<u>0.02$\pm$.00</u>
	20	0.34 $\pm$.02	0.04$\pm$.01	0.24 $\pm$.02	<u>0.06$\pm$.01</u>
	50	1.90 $\pm$.01	<u>0.54$\pm$.05</u>	1.90 $\pm$.01	0.25$\pm$.03
	100	1.95 $\pm$.00	<u>0.98$\pm$.04</u>	1.21 $\pm$.01	0.59$\pm$.04
2000	10	0.03 $\pm$.00	0.01$\pm$.00	0.02 $\pm$.00	<u>0.02$\pm$.00</u>
	20	0.14 $\pm$.01	0.01$\pm$.00	0.12 $\pm$.01	<u>0.03$\pm$.00</u>
	50	1.90 $\pm$.01	<u>0.44$\pm$.05</u>	1.83 $\pm$.01	0.10$\pm$.02
	100	1.95 $\pm$.00	<u>0.72$\pm$.05</u>	1.08 $\pm$.01	0.28$\pm$.03
5000	10	0.01 $\pm$.00	0.00$\pm$.00	<u>0.01$\pm$.00</u>	0.02 $\pm$.00
	20	0.04 $\pm$.00	0.01$\pm$.00	<u>0.04$\pm$.00</u>	0.01 $\pm$.00
	50	1.90 $\pm$.01	<u>0.37$\pm$.05</u>	1.75 $\pm$.03	0.04$\pm$.01
	100	1.95 $\pm$.00	<u>0.46$\pm$.05</u>	0.80 $\pm$.02	0.14$\pm$.03

Subspace recovery accuracy. Table 3 reports the squared subspace distance m^2 aggregated across all scenarios. At low-to-moderate dimensions ($p \in \{10, 20\}$), RMAVE achieves the lowest error, though SMAVE remains competitive with gaps of at most 0.03 and both methods achieving near-optimal recovery ($m^2 \leq 0.06$). At $p \in \{50, 100\}$, the ranking reverses: SMAVE reduces error by 2–9 $\times$ compared to RMAVE, which degrades because its OPG initialization suffers from sparse neighborhoods in high-dimensional ambient space.

SMAVE’s robustness at high dimensions stems from (i) k -NN neighborhoods that adapt to local data density regardless of ambient dimension, and (ii) stochastic optimization that can escape poor initializations. The ablation against OPG-kNN confirms that k -NN localization alone is insufficient: OPG-kNN performs no better than OPG at $p = 50$ and only marginally improves at $p = 100$, far from SMAVE’s

Table 4. Runtime in seconds (mean over 10 scenarios $\times$ 10 replications). **Bold**: best; underline: second best.

n	p	OPG	RMAVE	OPG-kNN	SMAVE
1000	10	<u>0.09</u>	0.79	0.01	0.20
	20	<u>0.13</u>	1.27	0.02	0.17
	50	0.25	3.18	0.05	<u>0.24</u>
	100	0.57	4.70	0.12	<u>0.41</u>
2000	10	0.29	2.67	0.03	<u>0.23</u>
	20	0.45	3.82	0.04	<u>0.22</u>
	50	0.91	11.67	0.10	<u>0.33</u>
	100	2.08	17.74	0.24	<u>0.61</u>
5000	10	1.67	15.54	0.08	<u>0.38</u>
	20	2.51	16.52	0.14	<u>0.46</u>
	50	5.57	54.27	0.30	<u>0.82</u>
	100	12.30	80.00	0.65	<u>1.67</u>


 Figure 1. Accuracy–efficiency trade-off at high dimensions ($p \in \{50, 100\}$, $n \in \{2000, 5000\}$). SMAVE consistently occupies the favorable lower-left region.

accuracy. The critical difference is operating in the projected space $\mathbb{R}^d$ rather than the ambient space $\mathbb{R}^p$.

Figure 2 (Appendix D.3) illustrates these dynamics. RMAVE starts with lower error but plateaus early, while SMAVE continues improving from random initialization throughout the 100 iterations. The wider confidence bands for SMAVE reflect stochastic gradient variance, yet this variability proves beneficial for escaping suboptimal basins that trap RMAVE’s deterministic updates.

Accuracy–runtime tradeoff. Table 4 reveals the cost of statistical accuracy. RMAVE achieves strong estimates but its $O(Tn^2p)$ complexity becomes prohibitive at scale. OPG-kNN is fastest due to its closed-form solution and sparse weights, but sacrifices accuracy. SMAVE resolves this trade-off: it runs much faster than RMAVE while achieving superior accuracy at high dimensions. Figure 1 illustrates this trade-off.

Table 5. Real-world datasets for prediction experiments.

Dataset	n	p	d candidates
Wine Quality	6,497	11	$\{2, 4, 6, 8\}$
Seoul Bike	8,760	14	$\{2, 4, 6, 8, 10\}$
Pumadyn	8,192	32	$\{2, 4, 8, 12, 16\}$
Gas Sensor	2,565	128	$\{5, 10, 20, 40, 60\}$

5.3. Real-data experiments

We evaluate SDR methods on real regression tasks where the true subspace is unknown and performance is measured by downstream prediction accuracy.

Datasets. Table 5 summarizes five publicly available datasets spanning diverse domains and scales. **Wine Quality** (Cortez et al., 2009) combines red and white wine samples, predicting sensory quality scores from 11 physico-chemical properties. **Seoul Bike** (Sathishkumar et al., 2020) predicts hourly rental demand from weather conditions and temporal features. **Pumadyn** (Orr et al., 2000) predicts angular acceleration from 32 attributes of simulated robot arm dynamics. **Gas Sensor** (Fonollosa et al., 2015) predicts gas concentration from a 128-dimensional metal-oxide sensor array, providing a high-dimensional test case where the curse of dimensionality is most pronounced.

Evaluation protocol. We perform 5-fold cross-validation repeated 3 times, yielding 15 train/test splits per configuration. All features are standardized to zero mean and unit variance on the training fold; the same transformation is applied to the test fold. For each candidate reduced dimension d , we apply the SDR method to training data, project both splits via $Z = X\hat{\mathbf{B}}$, and fit a distance-weighted k -NN regressor with $k = \min(10, \lfloor n_{\text{train}}/5 \rfloor)$ on the projected training data. We select $d^* = \arg \min_d \text{MSE}_{\text{CV}}(d)$ based on mean test MSE across all 15 splits and report the corresponding performance.

The candidate dimensions d in Table 5 are chosen to span a range from aggressive reduction to near-full rank, scaled to each dataset’s ambient dimension p . For low-dimensional datasets ($p \leq 15$), candidates cover most of the range up to $p - 1$. For moderate dimensions ($p \in [26, 32]$), candidates extend to roughly $p/2$. For the high-dimensional Gas Sensor dataset ($p = 128$), candidates range from 5 to 60, testing whether SDR methods can identify a low-dimensional structure amid many irrelevant features. This design ensures fair comparison: all methods search over the same candidate set, and the best d^* is selected independently for each method.

We compare OPG, OPG-kNN, RMAVE, and SMAVE from Section 3, along with PCA as an unsupervised baseline. RMAVE and SMAVE run for $T = 100$ iterations. SMAVE hyperparameters are fixed to the values tuned on synthetic

Table 6. Test MSE (mean $\pm$ s.d. over 15 splits) and selected dimension d^* . **Bold**: best MSE; underline: second best.

Dataset	Method	MSE	d^*	Time (s)
Wine Quality $n=6497$ $p=11$	PCA	0.515 ± 0.015	8	< 0.1
	OPG	0.521 ± 0.018	8	1.4
	OPG-kNN	0.521 ± 0.024	8	< 0.1
	RMAVE	0.498 ± 0.017	8	600.2
	SMAVE	<u>0.511 ± 0.018</u>	8	0.2
Seoul Bike $n=8760$ $p=14$	PCA	0.215 ± 0.015	10	< 0.1
	OPG	0.268 ± 0.012	8	2.8
	OPG-kNN	0.270 ± 0.012	8	0.1
	RMAVE	0.165 ± 0.008	4	327.3
	SMAVE	<u>0.167 ± 0.011</u>	4	0.3
Pumadyn $n=8192$ $p=32$	PCA	1.070 ± 0.048	16	< 0.1
	OPG	0.174 ± 0.009	2	4.4
	OPG-kNN	0.240 ± 0.020	2	0.3
	RMAVE	0.092 ± 0.003	2	55.4
	SMAVE	<u>0.127 ± 0.014</u>	4	0.5
Gas Sensor $n=2565$ $p=128$	PCA	0.020 ± 0.011	10	< 0.1
	OPG	0.026 ± 0.014	20	1.9
	OPG-kNN	0.025 ± 0.014	40	0.3
	RMAVE	0.025 ± 0.013	5	35.1
	SMAVE	0.018 ± 0.011	5	0.2

Table 7. Direct comparison of SMAVE and RMAVE for sufficient dimension reduction. Test MSE (mean $\pm$ std) computed over 15 independent cross-validation splits. The p -value column reports paired t -test results; bold marks the method with significantly lower error ($p < 0.05$).

Dataset	RMAVE	SMAVE	p -value
Wine Quality	0.498 ± 0.017	0.511 ± 0.018	9.2×10^{-5}
Seoul Bike	0.165 ± 0.008	0.167 ± 0.011	0.47
Pumadyn	0.092 ± 0.003	0.127 ± 0.014	9.4×10^{-8}
Gas Sensor	0.025 ± 0.013	0.018 ± 0.011	0.027

data (Appendix C), with no dataset-specific adjustments.

Results. Figure 3 (Appendix E) shows test MSE across all candidate dimensions; Table 6 reports results at the optimal d^* selected independently per method by cross-validation. Test MSE is computed on standardized responses to enable cross-dataset comparison. RMAVE runtime excludes its OPG initialization.

Analysis. Table 7 provides a direct comparison between SMAVE and RMAVE. RMAVE achieves significantly lower error on Wine Quality ($p = 9.2 \times 10^{-5}$) and Pumadyn ($p = 9.4 \times 10^{-8}$), though the absolute differences are modest (0.013 and 0.035, respectively) while runtime differs by 2–3 orders of magnitude. On Seoul Bike, the two methods are statistically indistinguishable ($p = 0.47$). On Gas Sensor ($p = 128$), SMAVE significantly outperforms RMAVE ($p = 0.027$), suggesting the stochastic approach scales better in high dimensions.

Both MAVE-based methods consistently select more parsimonious subspaces than OPG variants: $d^* = 5$ on Gas Sensor versus $d^* = 20$ –40 for OPG and OPG-kNN, achieving lower error with 4 – $8 \times$ fewer dimensions. This pattern suggests the MAVE objective better identifies the minimal sufficient subspace.

Finally, OPG and OPG-kNN yield nearly identical results across all datasets, indicating that k -NN localization alone does not improve OPG. The gains of SMAVE thus stem from the combination of projected-space localization and stochastic Riemannian optimization, not from either component in isolation.

6. Conclusion

We introduced SMAVE, a scalable algorithm for sufficient dimension reduction that combines Riemannian stochastic optimization on the Stiefel manifold with adaptive k -NN localization in the projected space. Our theoretical analysis established that minimizers of the MAVE objective recover the gradient outer product subspace—the same target as OPG—but through local regression rather than gradient aggregation, with approximation error $O(u^2)$ as the localization scale $u \rightarrow 0$.

Empirically, SMAVE achieves 10 – $50 \times$ speedups over RMAVE on synthetic benchmarks while matching or exceeding accuracy, with particularly pronounced gains in high dimensions ($p \geq 50$) where OPG-based initialization degrades. On real regression tasks, these computational savings translate to comparable prediction accuracy at a fraction of the cost, with speedups exceeding three orders of magnitude on some datasets. The ablation comparing OPG-kNN and SMAVE confirms that the gains stem from the combination of projected-space localization and stochastic Riemannian optimization, not from either component alone.

Two theoretical questions remain open. First, establishing convergence guarantees for SMAVE requires extending Riemannian SGD theory to objectives with iterate-dependent neighborhoods—the periodic k -NN rebuilding violates standard smoothness assumptions. Second, deriving finite-sample rates for the MAVE estimator would connect our population-level analysis to statistical guarantees. We view these as interesting directions for future work at the intersection of manifold optimization and nonparametric statistics.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Absil, P.-A., Mahony, R., and Sepulchre, R. *Optimization algorithms on matrix manifolds*. Princeton University Press, 2008.
- Adraghi, K. P. Minimum average deviance estimation for sufficient dimension reduction. *Journal of Statistical Computation and Simulation*, 88(3):411–431, 2018.
- Ausset, G., Cl  men  on, S., and Portier, F. Nearest neighbour based estimates of gradients: Sharp nonasymptotic bounds and applications. In *AISTATS proceedings*, pp. 532–540. PMLR, 2021.
- B  cigneul, G. and Ganeva, O.-E. Riemannian adaptive optimization methods. *arXiv preprint arXiv:1810.00760*, 2018.
- Bonnabel, S. Stochastic gradient descent on riemannian manifolds. *IEEE Transactions on Automatic Control*, 58(9):2217–2229, 2013.
- Boumal, N. *An introduction to optimization on smooth manifolds*. Cambridge University Press, 2023.
- Chen, S., Garcia, A., Hong, M., and Shahrampour, S. Decentralized riemannian gradient descent on the stiefel manifold. In *International Conference on Machine Learning*, pp. 1594–1605. PMLR, 2021.
- Cook, R. D. and Weisberg, S. Sliced inverse regression for dimension reduction: Comment. *Journal of the American Statistical Association*, 86(414):328–332, 1991.
- Cortez, P., Cerdeira, A., Almeida, F., Matos, T., and Reis, J. Modeling wine preferences by data mining from physicochemical properties. *Decision support systems*, 47(4):547–553, 2009.
- Dalalyan, A. S. and Spokoiny, V. G. Structural adaptive estimation for single-index models. *Annals of Statistics*, 36(3):1400–1436, 2008. doi: 10.1214/07-AOS516.
- Dalalyan, A. S., Juditsky, A., and Spokoiny, V. A new algorithm for estimating the effective dimension-reduction subspace. *Journal of Machine Learning Research*, 9(53):1647–1678, 2008. URL <http://jmlr.org/papers/v9/dalalyan08a.html>.
- Edelman, A., Arias, T. A., and Smith, S. T. The geometry of algorithms with orthogonality constraints. *SIAM journal on Matrix Analysis and Applications*, 20(2):303–353, 1998.
- Fonollosa, J., Sheik, S., Huerta, R., and Marco, S. Reservoir computing compensates slow response of chemosensor arrays exposed to fast varying gas concentrations in continuous monitoring. *Sensors and Actuators B: Chemical*, 215:618–629, 2015.
- Fukumizu, K., Bach, F. R., and Jordan, M. I. Kernel dimension reduction in regression. *Annals of Statistics*, 37(4):1871–1905, 2009a. doi: 10.1214/08-AOS637.
- Fukumizu, K., Bach, F. R., and Jordan, M. I. Kernel dimension reduction in regression. 2009b.
- Ghosh, B., Crowley, M., Karray, F., and Ghodsi, A. Spectral metric learning. In *Elements of Dimensionality Reduction and Manifold Learning*, pp. 313–351. Springer, 2022.
- H  rdle, W. and Stoker, T. M. Investigating smooth multiple regression by the method of average derivatives. *Journal of the American Statistical Association*, 84(408):986–995, 1989.
- Hristache, M., Juditsky, A., and Spokoiny, V. Direct estimation of the index coefficients in a single-index model. *Annals of Statistics*, 29(3):595–623, 2001. doi: 10.1214/aos/1009210686.
- Kulis, B. et al. Metric learning: A survey. *Foundations and Trends   in Machine Learning*, 5(4):287–364, 2013.
- Li, B. and Wang, S. On directional regression for dimension reduction. *Journal of the American Statistical Association*, 102(479):997–1008, 2007.
- Li, K.-C. Sliced inverse regression for dimension reduction. *Journal of the American Statistical Association*, 86(414):316–327, 1991.
- Li, L. Sparse sufficient dimension reduction. *Biometrika*, 94(3):603–613, 2007.
- Orr, M., Hallam, J., Murray, A., and Leonard, T. Assessing rbf networks using delve. *International Journal of Neural Systems*, 10(05):397–415, 2000.
- Park, J., Ahn, J., and Park, C. Kernel sufficient dimension reduction and variable selection for compositional data via amalgamation. In Krause, A., Brunskill, E., Cho, K., Engelhardt, B., Sabato, S., and Scarlett, J. (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 27034–27047. PMLR, 23–29 Jul 2023.

- Park, J., Kang, M., Lee, S., Lee, H., Kim, S., and Lee, J. Riemannian optimization for lora on the stiefel manifold. *arXiv preprint arXiv:2508.17901*, 2025.
- Sathishkumar, V. E., Park, J., and Cho, Y. Using data mining techniques for bike sharing demand prediction in metropolitan city. *Computer Communications*, 153: 353–366, 2020.
- Silverman, B. W. *Density estimation for statistics and data analysis*. Routledge, 2018.
- Wu, C., Miller, J., Chang, Y., Sznaiier, M., and Dy, J. Solving interpretable kernel dimensionality reduction. *Advances in neural information processing systems*, 32, 2019.
- Xia, Y. A constructive approach to the estimation of dimension reduction directions. *The Annals of Statistics*, pp. 2654–2690, 2007.
- Xia, Y., Tong, H., Li, W. K., and Zhu, L.-X. An adaptive estimation of dimension reduction space. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 64(3):363–410, 2002.
- Zhang, H., J Reddi, S., and Sra, S. Riemannian svrg: Fast stochastic optimization on riemannian manifolds. *Advances in Neural Information Processing Systems*, 29, 2016.

A. Riemannian geometry background

We provide the necessary background on Riemannian optimization over the Stiefel manifold. For a comprehensive treatment, we refer the reader to [Absil et al. \(2008\)](#); [Boumal \(2023\)](#).

The Stiefel and Grassmann manifolds. The Stiefel manifold $\text{St}(p, d)$ is the set of $p \times d$ matrices with orthonormal columns:

$$\text{St}(p, d) = \{\mathbf{B} \in \mathbb{R}^{p \times d} : \mathbf{B}^\top \mathbf{B} = I_d\}.$$

It is a compact embedded submanifold of $\mathbb{R}^{p \times d}$ of dimension $pd - d(d+1)/2$.

The Grassmann manifold $\text{Gr}(p, d)$ is the set of all d -dimensional linear subspaces of $\mathbb{R}^p$. Since two matrices $\mathbf{B}, \mathbf{B}' \in \text{St}(p, d)$ span the same subspace if and only if $\mathbf{B}' = \mathbf{B}\mathbf{Q}$ for some orthogonal matrix $\mathbf{Q} \in \mathcal{O}(d)$, the Grassmann manifold can be identified with the quotient

$$\text{Gr}(p, d) \cong \text{St}(p, d) / \mathcal{O}(d).$$

Optimizing over $\text{St}(p, d)$ as a convenient computational representation, with the understanding that any two solutions differing by a right orthogonal factor represent the same subspace.

Tangent space and Riemannian metric. We equip $\text{St}(p, d)$ with the Riemannian metric induced by the Frobenius inner product $\langle A, C \rangle = \text{tr}(A^\top C)$. The tangent space to $\text{St}(p, d)$ at $\mathbf{B}$ is

$$T_{\mathbf{B}}\text{St}(p, d) = \{\xi \in \mathbb{R}^{p \times d} : \mathbf{B}^\top \xi + \xi^\top \mathbf{B} = 0\},$$

which consists of matrices whose product with $\mathbf{B}^\top$ is skew-symmetric. The orthogonal projection of $Z \in \mathbb{R}^{p \times d}$ onto $T_{\mathbf{B}}\text{St}(p, d)$ is given by

$$\Pi_{T_{\mathbf{B}}\text{St}}(Z) = Z - \mathbf{B} \text{sym}(\mathbf{B}^\top Z), \quad \text{where } \text{sym}(A) = \frac{1}{2}(A + A^\top).$$

Riemannian gradient. For a smooth function $f : \text{St}(p, d) \rightarrow \mathbb{R}$, the Riemannian gradient $\text{grad } f(\mathbf{B}) \in T_{\mathbf{B}}\text{St}(p, d)$ is the unique tangent vector satisfying

$$df(\mathbf{B})[\xi] = \langle \text{grad } f(\mathbf{B}), \xi \rangle \quad \text{for all } \xi \in T_{\mathbf{B}}\text{St}(p, d).$$

Since $\text{St}(p, d)$ is an embedded submanifold of $\mathbb{R}^{p \times d}$, the Riemannian gradient is obtained by projecting the Euclidean gradient onto the tangent space:

$$\text{grad } f(\mathbf{B}) = \Pi_{T_{\mathbf{B}}\text{St}}(\nabla f(\mathbf{B})).$$

Retractions. The exponential map $\exp_x : T_x \mathcal{M} \rightarrow \mathcal{M}$ sends a tangent vector ξ to the endpoint of the geodesic starting at x with initial velocity ξ , providing the canonical way to move from a point along a tangent direction while staying on the manifold. A retraction $R : T\mathcal{M} \rightarrow \mathcal{M}$ relaxes this by requiring only first-order agreement with the exponential map: (i) $R_x(0_x) = x$, and (ii) $\frac{d}{dt}\big|_{t=0} R_x(t\xi) = \xi$ for all $\xi \in T_x \mathcal{M}$. These conditions suffice for convergence of gradient-based methods while being cheaper to compute. On the Stiefel manifold, we use the QR retraction, which approximates the exponential map at the first order. Given a matrix $A \in \mathbb{R}^{p \times d}$ with full column rank, let $A = QR$ be its thin QR decomposition, where $Q \in \text{St}(p, d)$ and $R \in \mathbb{R}^{d \times d}$ is upper triangular with positive diagonal entries. We denote by $\text{qf}(A) = Q$ the Q-factor of this decomposition. The QR retraction is then defined as

$$R_{\mathbf{B}}^{\text{QR}}(\xi) = \text{qf}(\mathbf{B} + \xi), \quad \xi \in T_{\mathbf{B}}\text{St}(p, d),$$

which maps the tangent vector ξ to a point on the manifold by orthonormalizing the columns of $\mathbf{B} + \xi$. This retraction has computational complexity $O(pd^2)$.

B. Proofs

Discussion on assumptions Recall some details about the mathematical background of the paper. Let (Y, X) be a random vector where $Y \in \mathbb{R}$ and $X \in \mathbb{R}^p$. We consider the regression function $g(x) := \mathbb{E}[Y \mid X = x]$ and write $Y = g(X) + \varepsilon$ where $\mathbb{E}[\varepsilon \mid X] = 0$ and $\text{var}(\varepsilon \mid X) = \sigma^2(X)$.

For a matrix $B \in \mathbb{R}^{p \times d}$ with orthonormal columns (i.e., $B \in \text{St}(p, d)$), we denote by $P_B = BB^\top$ the orthogonal projection onto the column space of B .

For $x \in \mathbb{R}^p$ and $u > 0$, let $\mathbb{E}_{(x,u)}[\cdot]$ denote the conditional expectation given that $X \in B(x, u) = \{y \in \mathbb{R}^p : \|y - x\| \leq u\}$. The local objective function is defined as:

$$D_{(x,u)}(B) := \min_{a \in \mathbb{R}, b \in \mathbb{R}^d} \mathbb{E}_{(x,u)} [(Y - a - b^\top B^\top (X - x))^2]. \quad (11)$$

Assumption B.1 (L -smoothness). There exists $L > 0$ such that, for u small enough,

$$\mathbb{E}_{(x,u)} [|g(X) - g(x) - \nabla g(x)^\top (X - x)|^2] \leq \frac{L^2}{4} u^4. \quad (12)$$

Assumption B.2 (Local covariance lower bound). There exists $\lambda \in (0, 1]$ such that, for u small enough,

$$G_{(x,u)} \succeq \lambda u^2 I_p, \quad (13)$$

where $G_{(x,u)} := \mathbb{E}_{(x,u)} [(X - \mathbb{E}_{(x,u)}[X])(X - \mathbb{E}_{(x,u)}[X])^\top]$ is the local covariance matrix.

On the constraint $\lambda \leq 1$: For any real random variable Y and any fixed $y \in \mathbb{R}$, the mean minimizes the expected squared deviation: $\mathbb{E}[(Y - \mathbb{E}[Y])^2] \leq \mathbb{E}[(Y - y)^2]$. Applying this to $Y = X^\top v$ for arbitrary $v \in \mathbb{R}^p$ and the fixed point $x^\top v$ yields $v^\top G_{(x,u)} v \leq v^\top \mathbb{E}_{(x,u)} [(X - x)(X - x)^\top] v$. Since this holds for all v , we obtain the matrix inequality $G_{(x,u)} \preceq \mathbb{E}_{(x,u)} [(X - x)(X - x)^\top]$. Under $\mathbb{E}_{(x,u)}$, we have $X \in B(x, u)$, so $\|X - x\| \leq u$ almost surely, which implies $\mathbb{E}_{(x,u)} [(X - x)(X - x)^\top] \preceq u^2 I_p$. Combining these inequalities: $G_{(x,u)} \preceq u^2 I_p$, hence $\lambda \leq 1$.

This section contains the proofs of the mathematical statements of the paper.

B.1. Proof of Proposition 2.4

Proof of upper bound. We need to show that if $(I_p - P_B)\nabla g(x) = 0$ (i.e., $\nabla g(x) \in \text{span}(B)$), then

$$D_{(x,u)}(B) \leq \mathbb{E}_{(x,u)} [\sigma^2(X)] + \frac{L^2}{4} u^4. \quad (14)$$

Since $Y = g(X) + \varepsilon$ with $\mathbb{E}[\varepsilon | X] = 0$, we have the decomposition:

$$\mathbb{E}_{(x,u)} [(Y - a - b^\top B^\top (X - x))^2] = \mathbb{E}_{(x,u)} [\sigma^2(X)] + \mathbb{E}_{(x,u)} [(g(X) - a - b^\top B^\top (X - x))^2]. \quad (15)$$

Indeed, expanding the square and using $\mathbb{E}_{(x,u)} [\varepsilon \cdot h(X)] = \mathbb{E}_{(x,u)} [h(X) \cdot \mathbb{E}[\varepsilon | X]] = 0$ for any function h .

Thus, minimizing over (a, b) gives:

$$D_{(x,u)}(B) = \mathbb{E}_{(x,u)} [\sigma^2(X)] + \min_{a \in \mathbb{R}, b \in \mathbb{R}^d} \mathbb{E}_{(x,u)} [(g(X) - a - b^\top B^\top (X - x))^2]. \quad (16)$$

Take $a = g(x)$ and $b = B^\top \nabla g(x)$. With this choice:

$$b^\top B^\top (X - x) = \nabla g(x)^\top B B^\top (X - x) = \nabla g(x)^\top P_B (X - x).$$

Now we use the gradient condition: If $(I_p - P_B)\nabla g(x) = 0$, then $P_B \nabla g(x) = \nabla g(x)$, so:

$$b^\top B^\top (X - x) = \nabla g(x)^\top (X - x).$$

By definition of the minimum:

$$\min_{a,b} \mathbb{E}_{(x,u)} [(g(X) - a - b^\top B^\top (X - x))^2] \leq \mathbb{E}_{(x,u)} [(g(X) - g(x) - \nabla g(x)^\top (X - x))^2] \leq \frac{L^2}{4} u^4,$$

where the last inequality uses Assumption B.1. □

Proof of lower bound: We need to show that whenever $uL \leq \frac{\lambda}{2} \|(I_p - P_B) \nabla g(x)\|_2$,

$$D_{(x,u)}(B) \geq \mathbb{E}_{(x,u)}[\sigma^2(X)] + \frac{\lambda u^2}{2} \|(I_p - P_B) \nabla g(x)\|_2^2. \quad (17)$$

We adopt the following strategy. We optimize with respect to a first, characterize the optimal b , then bound from below.

Step 1: Optimize over a . Define the centered quantities:

$$\tilde{g}_{(x,u)} := g(X) - \mathbb{E}_{(x,u)}[g(X)], \quad (18)$$

$$\tilde{X}_{(x,u)} := X - \mathbb{E}_{(x,u)}[X]. \quad (19)$$

Optimizing over a (taking $a^* = \mathbb{E}_{(x,u)}[g(X)] - b^\top B^\top \mathbb{E}_{(x,u)}[X - x]$):

$$\min_{a \in \mathbb{R}} \mathbb{E}_{(x,u)} [(g(X) - a - b^\top B^\top (X - x))^2] = \mathbb{E}_{(x,u)} [(\tilde{g}_{(x,u)} - b^\top B^\top \tilde{X}_{(x,u)})^2]. \quad (20)$$

Step 2: Characterize the optimal b . After optimizing over a , we need to solve:

$$\min_{b \in \mathbb{R}^d} L(b) := \mathbb{E}_{(x,u)} [(\tilde{g}_{(x,u)} - b^\top B^\top \tilde{X}_{(x,u)})^2], \quad (21)$$

where $B \in \text{St}(p, d)$ is fixed.

Step 2.1: Expand the objective. Expanding the square:

$$L(b) = \mathbb{E}_{(x,u)}[\tilde{g}_{(x,u)}^2] - 2b^\top B^\top \mathbb{E}_{(x,u)}[\tilde{X}_{(x,u)} \tilde{g}_{(x,u)}] + b^\top B^\top G_{(x,u)} B b.$$

Define $\mu_{(x,u)} := \mathbb{E}_{(x,u)}[\tilde{X}_{(x,u)} \tilde{g}_{(x,u)}] \in \mathbb{R}^p$ and $H := B^\top G_{(x,u)} B \in \mathbb{R}^{d \times d}$. Then:

$$L(b) = \mathbb{E}_{(x,u)}[\tilde{g}_{(x,u)}^2] - 2b^\top B^\top \mu_{(x,u)} + b^\top H b. \quad (22)$$

Step 2.2: Solve the first-order condition. Setting $\nabla_b L(b) = 0$:

$$B^\top G_{(x,u)} B \cdot b = B^\top \mu_{(x,u)}.$$

We verify that $H = B^\top G_{(x,u)} B$ is invertible. By Assumption B.2, $G_{(x,u)} \succeq \lambda u^2 I_p$. Since $B \in \text{St}(p, d)$ has orthonormal columns, for any $v \in \mathbb{R}^d$:

$$v^\top H v = v^\top B^\top G_{(x,u)} B v = (Bv)^\top G_{(x,u)} (Bv) \geq \lambda u^2 \|Bv\|_2^2 = \lambda u^2 \|v\|_2^2,$$

where the last equality uses $B^\top B = I_d$. Hence $H \succeq \lambda u^2 I_d$, guaranteeing invertibility.

The unique solution is:

$$b_{(x,u)} = (B^\top G_{(x,u)} B)^{-1} B^\top \mathbb{E}_{(x,u)}[\tilde{X}_{(x,u)} \tilde{g}_{(x,u)}]. \quad (23)$$

Step 3: Quadratic expansion. Define the linearization residual:

$$L_{(x,u)}^* := \tilde{g}_{(x,u)} - \nabla g(x)^\top \tilde{X}_{(x,u)}. \quad (24)$$

We can write:

$$\tilde{g}_{(x,u)} - b_{(x,u)}^\top B^\top \tilde{X}_{(x,u)} = L_{(x,u)}^* + (\nabla g(x) - B b_{(x,u)})^\top \tilde{X}_{(x,u)}.$$

Expanding the square:

$$\mathbb{E}_{(x,u)} [(\tilde{g}_{(x,u)} - b_{(x,u)}^\top B^\top \tilde{X}_{(x,u)})^2] \quad (25)$$

$$= \mathbb{E}_{(x,u)} [(L_{(x,u)}^*)^2] + 2\mathbb{E}_{(x,u)} [L_{(x,u)}^* \cdot (\nabla g(x) - B b_{(x,u)})^\top \tilde{X}_{(x,u)}] + \|\nabla g(x) - B b_{(x,u)}\|_{G_{(x,u)}}^2, \quad (26)$$

where $\|v\|_{G(x,u)}^2 := v^\top G(x,u)v$.

Step 4: Bound the first term. By Assumption B.1:

$$\mathbb{E}_{(x,u)}[(L_{(x,u)}^*)^2] \leq \frac{L^2}{4}u^4. \quad (27)$$

Step 5: Bound the cross term. Let $v := \nabla g(x) - Bb_{(x,u)}$. By the Cauchy–Schwarz inequality:

$$\left| \mathbb{E}_{(x,u)}[L_{(x,u)}^* \cdot v^\top \tilde{X}_{(x,u)}] \right| \leq \sqrt{\mathbb{E}_{(x,u)}[(L_{(x,u)}^*)^2]} \cdot \sqrt{\mathbb{E}_{(x,u)}[(v^\top \tilde{X}_{(x,u)})^2]}.$$

For the first factor, by (27):

$$\sqrt{\mathbb{E}_{(x,u)}[(L_{(x,u)}^*)^2]} \leq \frac{L}{2}u^2.$$

For the second factor, note that $\mathbb{E}_{(x,u)}[\|\tilde{X}_{(x,u)}\|_2^2] \leq \mathbb{E}_{(x,u)}[\|X - x\|_2^2] \leq u^2$ (since $X \in B(x, u)$ under $\mathbb{E}_{(x,u)}$, and by the variance-minimization property: the covariance is bounded by the second moment about any fixed point). Thus:

$$\mathbb{E}_{(x,u)}[(v^\top \tilde{X}_{(x,u)})^2] \leq \|v\|_2^2 \cdot \mathbb{E}_{(x,u)}[\|\tilde{X}_{(x,u)}\|_2^2] \leq u^2\|v\|_2^2.$$

Combining:

$$\left| \mathbb{E}_{(x,u)}[L_{(x,u)}^* \cdot v^\top \tilde{X}_{(x,u)}] \right| \leq \frac{L}{2}u^2 \cdot u\|v\|_2 = \frac{L}{2}u^3\|v\|_2. \quad (28)$$

Step 6: Lower bound on the quadratic term. Since $Bb_{(x,u)} \in \text{span}(B)$, by the Pythagorean theorem:

$$\|\nabla g(x) - Bb_{(x,u)}\|_2^2 = \|(I_p - P_B)\nabla g(x)\|_2^2 + \|P_B\nabla g(x) - Bb_{(x,u)}\|_2^2 \geq \|(I_p - P_B)\nabla g(x)\|_2^2. \quad (29)$$

By Assumption B.2:

$$\|\nabla g(x) - Bb_{(x,u)}\|_{G(x,u)}^2 \geq \lambda u^2 \|\nabla g(x) - Bb_{(x,u)}\|_2^2 \geq \lambda u^2 \|(I_p - P_B)\nabla g(x)\|_2^2. \quad (30)$$

Step 7: Combine the bounds. Let $\delta := \|(I_p - P_B)\nabla g(x)\|_2$. From (25), dropping the non-negative term $\mathbb{E}_{(x,u)}[(L_{(x,u)}^*)^2]$ and using (28) and (30):

$$\begin{aligned} \mathbb{E}_{(x,u)} \left[(\tilde{g}_{(x,u)} - b_{(x,u)}^\top B^\top \tilde{X}_{(x,u)})^2 \right] &\geq \lambda u^2 \|v\|_2^2 - 2 \cdot \frac{L}{2}u^3\|v\|_2 \\ &= \lambda u^2 \|v\|_2^2 - Lu^3\|v\|_2 \\ &= u^2\|v\|_2 (\lambda\|v\|_2 - Lu). \end{aligned}$$

Since $\|v\|_2 \geq \delta$ by (29), whenever $\lambda\delta \geq Lu$ (equivalently, $uL \leq \lambda\delta$), we have:

$$\lambda\|v\|_2 - Lu \geq \lambda\delta - Lu \geq 0.$$

To obtain the coefficient $\lambda/2$ in the final bound, we need:

$$\lambda u^2\|v\|_2^2 - Lu^3\|v\|_2 \geq \frac{\lambda u^2}{2}\|v\|_2^2.$$

This simplifies to:

$$\frac{\lambda u^2}{2}\|v\|_2 \geq Lu^3 \iff \|v\|_2 \geq \frac{2L}{\lambda}u.$$

Since $\|v\|_2 \geq \delta$, a sufficient condition is $\delta \geq \frac{2L}{\lambda}u$, i.e.,

$$uL \leq \frac{\lambda}{2}\delta = \frac{\lambda}{2}\|(I_p - P_B)\nabla g(x)\|_2.$$

Under this condition:

$$\mathbb{E}_{(x,u)} \left[(\tilde{g}_{(x,u)} - b_{(x,u)}^\top B^\top \tilde{X}_{(x,u)})^2 \right] \geq \frac{\lambda u^2}{2} \|v\|_2^2 \geq \frac{\lambda u^2}{2} \delta^2 = \frac{\lambda u^2}{2} \|(I_p - P_B)\nabla g(x)\|_2^2.$$

Combining with the bias-variance decomposition (16):

$$D_{(x,u)}(B) \geq \mathbb{E}_{(x,u)}[\sigma^2(X)] + \frac{\lambda u^2}{2} \|(I_p - P_B)\nabla g(x)\|_2^2.$$

□

B.2. Proof of Proposition 2.5

Let $\mathbf{B}^*$ be a minimizer of $D_u(\mathbf{B}) = \int D_{(x,u)}(\mathbf{B}) \mathbb{P}(\mathrm{d}x)$. Since the central mean subspace has dimension at most d , there exists $\mathbf{B}_0 \in \text{St}(p, d)$ such that $\nabla g(x) \in \text{span}(\mathbf{B}_0)$ for $\mathbb{P}$ -almost every x . Define $w^*(x) := (I_p - P_{\mathbf{B}^*})\nabla g(x)$. *Step 1: Upper bound on $D_u(\mathbf{B}^*)$.* By optimality of $\mathbf{B}^*$ and Proposition 2.4(ii) applied to $\mathbf{B}_0$:

$$D_u(\mathbf{B}^*) \leq D_u(\mathbf{B}_0) \leq \int \mathbb{E}_{(x,u)}[\sigma^2(X)] \mathbb{P}(\mathrm{d}x) + \frac{L^2 u^4}{4}.$$

Step 2: Lower bound on $D_u(\mathbf{B}^)$.* Partition the support according to whether the condition in Proposition 2.4(i) holds:

$$A = \left\{ x : \|w^*(x)\|_2 \geq \frac{2Lu}{\lambda} \right\}, \quad A^c = \left\{ x : \|w^*(x)\|_2 < \frac{2Lu}{\lambda} \right\}.$$

On A , we have $u \leq \frac{\lambda}{2L}\|w^*(x)\|_2$, so Proposition 2.4(i) yields

$$D_{(x,u)}(\mathbf{B}^*) \geq \mathbb{E}_{(x,u)}[\sigma^2(X)] + \frac{\lambda u^2}{2} \|w^*(x)\|_2^2.$$

On A^c , we use the trivial bound $D_{(x,u)}(\mathbf{B}^*) \geq \mathbb{E}_{(x,u)}[\sigma^2(X)]$. Integrating over each region:

$$D_u(\mathbf{B}^*) = \int_A D_{(x,u)}(\mathbf{B}^*) \mathbb{P}(\mathrm{d}x) + \int_{A^c} D_{(x,u)}(\mathbf{B}^*) \mathbb{P}(\mathrm{d}x) \geq \int \mathbb{E}_{(x,u)}[\sigma^2(X)] \mathbb{P}(\mathrm{d}x) + \frac{\lambda u^2}{2} \int_A \|w^*(x)\|_2^2 \mathbb{P}(\mathrm{d}x).$$

Step 3: Combine bounds. Comparing Steps 1 and 2:

$$\frac{\lambda u^2}{2} \int_A \|w^*(x)\|_2^2 \mathbb{P}(\mathrm{d}x) \leq \frac{L^2 u^4}{4},$$

which gives $\int_A \|w^*(x)\|_2^2 \mathbb{P}(\mathrm{d}x) \leq \frac{L^2 u^2}{2\lambda}$. On A^c , by definition $\|w^*(x)\|_2^2 < \frac{4L^2 u^2}{\lambda^2}$, so

$$\int_{A^c} \|w^*(x)\|_2^2 \mathbb{P}(\mathrm{d}x) \leq \frac{4L^2 u^2}{\lambda^2} \mathbb{P}(A^c) \leq \frac{4L^2 u^2}{\lambda^2}.$$

Step 4: Conclude. Adding the contributions and using $\lambda \leq 1$:

$$E(\mathbf{B}^*, \mathcal{M}) = \int_A \|w^*(x)\|_2^2 \mathbb{P}(\mathrm{d}x) + \int_{A^c} \|w^*(x)\|_2^2 \mathbb{P}(\mathrm{d}x) \leq \frac{L^2 u^2}{2\lambda} + \frac{4L^2 u^2}{\lambda^2} \leq \frac{9L^2 u^2}{2\lambda^2}.$$

B.3. Proof of Proposition 2.7

Introduce the following notation. For each $j \in \{1, \dots, n\}$, define: the weighted means $\bar{Y}^{(j)} = \sum_{i=1}^n w_{ij} Y_i$ and $\bar{X}^{(j)} = \sum_{i=1}^n w_{ij} X_i$, and the weight matrix $W^{(j)} = \text{diag}(w_{1j}, \dots, w_{nj}) \in \mathbb{R}^{n \times n}$. The proof is based on the following lemma.

Lemma B.3 (Local regression subproblem). *For fixed $\mathbf{B} \in \text{St}(p, d)$ and $j \in \{1, \dots, n\}$, the local regression coefficients solving*

$$\min_{a, \mathbf{b}} \sum_{i=1}^n [Y_i - a - \mathbf{b}^\top \mathbf{B}^\top (X_i - X_j)]^2 w_{ij}$$

are given by

$$\begin{aligned} \hat{a}_j &= \bar{Y}^{(j)} - \hat{\mathbf{b}}_j^\top \mathbf{B}^\top (\bar{X}^{(j)} - X_j), \\ \hat{\mathbf{b}}_j &= (\mathbf{B}^\top G^{(j)} \mathbf{B})^* \mathbf{B}^\top \mu^{(j)}, \end{aligned}$$

where $(\cdot)^*$ denotes the Moore-Penrose pseudoinverse. Moreover, the corresponding residual sum of squares equals

$$\sigma_j^2(\mathbf{B}) := \sum_{i=1}^n [Y_i - \hat{a}_j - \hat{\mathbf{b}}_j^\top \mathbf{B}^\top (X_i - X_j)]^2 w_{ij} = \|(I_n - H^{(j)}) W^{(j)1/2} \tilde{Y}^{(j)}\|_2^2,$$

where $H^{(j)} = W^{(j)1/2} \tilde{Z}^{(j)} \mathbf{B} (\mathbf{B}^\top G^{(j)} \mathbf{B})^* \mathbf{B}^\top (\tilde{Z}^{(j)})^\top W^{(j)1/2}$ is the orthogonal projection onto the column space of $W^{(j)1/2} \tilde{Z}^{(j)} \mathbf{B}$.

Proof. Setting $\partial_a D_{n,j} = 0$ where $D_{n,j}(\mathbf{B}, a, \mathbf{b}) = \sum_{i=1}^n [Y_i - a - \mathbf{b}^\top \mathbf{B}^\top (X_i - X_j)]^2 w_{ij}$, and using $\sum_i w_{ij} = 1$, yields

$$a = \bar{Y}^{(j)} - \mathbf{b}^\top \mathbf{B}^\top (\bar{X}^{(j)} - X_j).$$

Substituting back, the problem reduces to the centered weighted least squares

$$\min_{\mathbf{b}} \sum_{i=1}^n \left\{ \tilde{Y}_i^{(j)} - \mathbf{b}^\top \mathbf{B}^\top \tilde{X}_i^{(j)} \right\}^2 w_{ij} = \min_{\mathbf{b}} \|W^{(j)1/2} (\tilde{Y}^{(j)} - \tilde{Z}^{(j)} \mathbf{B} \mathbf{b})\|_2^2,$$

which is a weighted least squares problem with design matrix $\tilde{Z}^{(j)} \mathbf{B} \in \mathbb{R}^{n \times d}$. The normal equations give the stated formula for $\hat{\mathbf{b}}_j$, and the minimum value is the squared norm of the residual after projection. $\square$

By Lemma B.3, after profiling out the local regression coefficients $(a_j, \mathbf{b}_j)$, the criterion (5) becomes

$$\min_{\mathbf{B} \in \text{St}(p, d)} \sum_{j=1}^n \sigma_j^2(\mathbf{B}) = \min_{\mathbf{B} \in \text{St}(p, d)} \sum_{j=1}^n \|(I_n - H^{(j)}) W^{(j)1/2} \tilde{Y}^{(j)}\|_2^2.$$

Since $H^{(j)}$ is an orthogonal projection, we have the decomposition

$$\|W^{(j)1/2} \tilde{Y}^{(j)}\|_2^2 = \|(I_n - H^{(j)}) W^{(j)1/2} \tilde{Y}^{(j)}\|_2^2 + \|H^{(j)} W^{(j)1/2} \tilde{Y}^{(j)}\|_2^2.$$

Since the first term on the left-hand side is constant with respect to $\mathbf{B}$, minimizing the residual sum of squares is equivalent to maximizing $\sum_{j=1}^n \|H^{(j)} W^{(j)1/2} \tilde{Y}^{(j)}\|_2^2$. Finally, a direct computation shows

$$\|H^{(j)} W^{(j)1/2} \tilde{Y}^{(j)}\|_2^2 = (\mu^{(j)})^\top \mathbf{B} (\mathbf{B}^\top G^{(j)} \mathbf{B})^* \mathbf{B}^\top \mu^{(j)},$$

which yields the result. For any orthogonal $\mathbf{Q}$,

$$(\mathbf{B} \mathbf{Q})^\top G^{(j)} (\mathbf{B} \mathbf{Q}) = \mathbf{Q}^\top (\mathbf{B}^\top G^{(j)} \mathbf{B}) \mathbf{Q},$$

and since the pseudoinverse satisfies $(\mathbf{Q}^\top \mathbf{M} \mathbf{Q})^* = \mathbf{Q}^\top \mathbf{M}^* \mathbf{Q}$ for orthogonal $\mathbf{Q}$, we have

$$\begin{aligned} & (\mathbf{B} \mathbf{Q})^\top \mu^{(j)} \left[(\mathbf{B} \mathbf{Q})^\top G^{(j)} (\mathbf{B} \mathbf{Q}) \right]^* (\mathbf{B} \mathbf{Q})^\top \mu^{(j)} \\ &= (\mu^{(j)})^\top \mathbf{B} \left(\mathbf{B}^\top G^{(j)} \mathbf{B} \right)^* \mathbf{B}^\top \mu^{(j)}. \end{aligned}$$

B.4. Proof of Proposition B.4

Proposition B.4 (Equivalence OPG with MAVE). *When $G^{(j)} = c \cdot I_p$ for all j and some constant $c > 0$, the MAVE and OPG estimators coincide. Both yield the top d eigenvectors of $M := \sum_{j=1}^n \mu^{(j)} (\mu^{(j)})^\top$.*

Proof. Under the assumption $G^{(j)} = c \cdot I_p$, the OPG gradient estimates simplify to $\hat{b}_j = c^{-1} \mu^{(j)}$, so

$$\hat{\Sigma}_{\text{OPG}} = \frac{1}{nc^2} \sum_{j=1}^n \mu^{(j)} (\mu^{(j)})^\top = \frac{1}{nc^2} M.$$

Thus, $\hat{\mathbf{B}}_{\text{OPG}}$ consists of the top d eigenvectors of M .

For MAVE, since $\mathbf{B}^\top G^{(j)} \mathbf{B} = c \cdot \mathbf{B}^\top \mathbf{B} = c \cdot I_d$ for $\mathbf{B} \in \text{St}(p, d)$, the objective (6) becomes

$$F(\mathbf{B}) = \frac{1}{c} \sum_{j=1}^n \left\| \mathbf{B}^\top \mu^{(j)} \right\|^2 = \frac{1}{c} \text{tr}(\mathbf{B}^\top M \mathbf{B}).$$

This is maximized by the top d eigenvectors of M . □

B.5. Proof of Proposition 4.1

We derive the Riemannian gradient of the MAVE objective (6). We first establish the gradient for a single summand.

Lemma B.5. *Let $G \in \mathbb{R}^{p \times p}$ be symmetric positive semi-definite and $\mu \in \mathbb{R}^p$. Define $f : \text{St}(p, d) \rightarrow \mathbb{R}$ by $f(\mathbf{B}) = \mu^\top \mathbf{B} (\mathbf{B}^\top G \mathbf{B})^{-1} \mathbf{B}^\top \mu$, assuming $\mathbf{B}^\top G \mathbf{B}$ is invertible. Then:*

1. *The Euclidean gradient is $\nabla f(\mathbf{B}) = 2(\mu - G\mathbf{B}u)u^\top$, where $u = (\mathbf{B}^\top G \mathbf{B})^{-1} \mathbf{B}^\top \mu$.*
2. *The Euclidean gradient already lies in the tangent space: $\nabla f(\mathbf{B}) \in T_{\mathbf{B}} \text{St}(p, d)$.*
3. *Consequently, $\text{grad } f(\mathbf{B}) = \nabla f(\mathbf{B})$.*

Proof. Let $M = \mathbf{B}^\top G \mathbf{B}$, $v = \mathbf{B}^\top \mu$, and $u = M^{-1}v$, so that $f(\mathbf{B}) = v^\top u$.

Part (i). Consider a perturbation $\Delta \in \mathbb{R}^{p \times d}$. The differentials are $dv = \Delta^\top \mu$ and $dM = \Delta^\top G \mathbf{B} + \mathbf{B}^\top G \Delta$. Using $d(M^{-1}) = -M^{-1}(dM)M^{-1}$, we compute

$$\begin{aligned} df &= 2(dv)^\top u + v^\top d(M^{-1})v \\ &= 2\mu^\top \Delta u - u^\top (\Delta^\top G \mathbf{B} + \mathbf{B}^\top G \Delta)u \\ &= 2\mu^\top \Delta u - 2u^\top \mathbf{B}^\top G \Delta u \\ &= 2\text{tr}(u(\mu - G\mathbf{B}u)^\top \Delta) \\ &= \langle 2(\mu - G\mathbf{B}u)u^\top, \Delta \rangle. \end{aligned}$$

Hence $\nabla f(\mathbf{B}) = 2(\mu - G\mathbf{B}u)u^\top$.

Part (ii). To verify that $\nabla f(\mathbf{B}) \in T_{\mathbf{B}} \text{St}(p, d)$, we check that $\mathbf{B}^\top \nabla f(\mathbf{B})$ is skew-symmetric:

$$\mathbf{B}^\top \nabla f(\mathbf{B}) = 2(\mathbf{B}^\top \mu - \mathbf{B}^\top G \mathbf{B}u)u^\top = 2(v - Mu)u^\top = 0,$$

since $Mu = v$ by definition of u . A zero matrix is trivially skew-symmetric.

Part (iii). Since $\nabla f(\mathbf{B}) \in T_{\mathbf{B}} \text{St}(p, d)$, the projection is the identity: $\text{grad } f(\mathbf{B}) = \Pi_{T_{\mathbf{B}} \text{St}}(\nabla f(\mathbf{B})) = \nabla f(\mathbf{B})$. □

Proposition 4.1 follows immediately by summing the contributions from each $j \in \{1, \dots, n\}$:

$$\text{grad } F(\mathbf{B}) = \sum_{j=1}^n \text{grad } f_j(\mathbf{B}) = 2 \sum_{j=1}^n \left(\mu^{(j)} - G^{(j)} \mathbf{B} u^{(j)} \right) (u^{(j)})^\top.$$

Remark B.6 (Computational simplification). The fact that $\nabla F(\mathbf{B})$ already lies in $T_{\mathbf{B}}\text{St}(p, d)$ eliminates the need for an explicit projection step in the gradient computation. This reduces per-iteration cost and simplifies implementation.

Remark B.7 (Regularization for singular cases). When $\mathbf{B}^\top G^{(j)} \mathbf{B}$ is singular or ill-conditioned, one may use the regularized objective

$$F_\epsilon(\mathbf{B}) = \sum_{j=1}^n (\mu^{(j)})^\top \mathbf{B} (\mathbf{B}^\top G^{(j)} \mathbf{B} + \epsilon I_d)^{-1} \mathbf{B}^\top \mu^{(j)}$$

for a small $\epsilon > 0$. The gradient formula extends with $u_\epsilon^{(j)} = (\mathbf{B}^\top G^{(j)} \mathbf{B} + \epsilon I_d)^{-1} \mathbf{B}^\top \mu^{(j)}$.

C. Implementation details

Shared settings. All kernel-based methods (OPG, RMAVE) use the Gaussian kernel $K_h(u) = \exp(-\|u\|^2/2h^2)$ with Silverman’s rule-of-thumb bandwidth (Silverman, 2018):

$$h = \hat{\sigma} \cdot n^{-1/(q+4)}, \quad (31)$$

where q is the operating dimension ($q = p$ for OPG, $q = d$ for RMAVE) and $\hat{\sigma}$ is the robust scale estimate (median absolute deviation across coordinates, scaled by $1.4826 = 1/\Phi^{-1}(3/4)$). For k -NN methods (OPG-kNN, SMAVE), we match the effective kernel neighborhood size $\sim nh^q$ by setting $k = \lceil n^{4/(q+4)} \rceil$, clipped to $[k_{\min}, n/3]$ with $k_{\min} = 50$ for OPG-kNN (operating in $\mathbb{R}^p$) and $k_{\min} = 20$ for SMAVE (operating in $\mathbb{R}^d$).

Both iterative methods run for $T = 100$ iterations. RMAVE is initialized with the OPG solution, while SMAVE is initialized by sampling a $p \times d$ matrix with i.i.d. $\mathcal{N}(0, 1)$ entries and orthonormalizing via QR decomposition, yielding a uniform draw on $\text{St}(p, d)$. All methods regularize matrix inversions with $\epsilon = 10^{-5}$. Implementations use Python with NumPy/SciPy; random seeds are fixed for reproducibility.

SMAVE hyperparameters. We tuned SMAVE hyperparameters via grid search over 7,560 synthetic configurations: five link functions $\times$ three sample sizes ($n \in \{1000, 2000, 5000\}$) $\times$ two dimensions ($p \in \{10, 20\}$) $\times$ five replications $\times$ 252 hyperparameter combinations. The tuning grid covered:

- Initial step size: $\alpha_0 \in \{0.05, 0.08, 0.1, 0.12, 0.15, 0.18, 0.2\}$
- Step size decay: $\gamma \in \{0.02, 0.04, 0.06\}$
- Momentum: $\beta \in \{0.0, 0.5, 0.9\}$
- k -NN refresh period: $\tau \in \{20, 25, 30, 50\}$

Analysis revealed that momentum hyperparameter $\beta = 0.9$ consistently reduced error by approximately $3\times$ compared to no momentum, making it the most impactful hyperparameter. Larger step sizes ($\alpha_0 \geq 0.15$) improved convergence within the 100-iteration budget. Lower k -NN refresh frequencies (20–30) yielded better accuracy but increased runtime, whereas $\tau = 50$ provided a 20% speedup with moderate accuracy loss. Balancing these trade-offs, we selected:

$$\alpha_0 = 0.2, \quad \gamma = 0.02, \quad \beta = 0.9, \quad \tau = 25.$$

The mini-batch size was set adaptively as $m = \min(200, \max(50, n/50))$. These hyperparameters, tuned exclusively on synthetic data, were held fixed for all real-data experiments.

D. Extended synthetic experiment results

D.1. Results by link functions

We detail the five considered link functions $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ that capture diverse nonlinear relationships:

$$\begin{aligned} g_1(\mathbf{z}) &= z_1(z_1 + z_2 + 1) && \text{(polynomial),} \\ g_2(\mathbf{z}) &= \sin(z_1) + \frac{1}{2} \cos(2z_2) && \text{(sinusoidal),} \\ g_3(\mathbf{z}) &= \exp(-z_1^2) + \frac{1}{2} z_2 && \text{(exponential),} \\ g_4(\mathbf{z}) &= z_1 z_2 + \sin(z_1) + \exp(-z_2^2/2) && \text{(interaction),} \\ g_5(\mathbf{z}) &= z_1 / (0.5 + (z_2 + 1)^2) && \text{(rational).} \end{aligned}$$

These functions span polynomial, periodic, localized, and rational nonlinearities. Each is paired with two covariance structures—identity ($\Sigma = \mathbf{I}_p$) and AR(1) ($\Sigma_{ij} = 0.5^{|i-j|}$)—yielding 10 scenarios.

Table 8 disaggregates performance by link function. SMAVE achieves the lowest error across all five scenarios, with particularly strong gains on symmetric nonlinearities (sinusoidal, exponential) where gradient magnitudes vary substantially across the input domain. The interaction and rational links yield the smallest errors overall, likely because their dominant gradient directions are more easily identifiable. OPG and OPG-kNN show little variation across links, confirming that their limiting factor is ambient-space localization rather than link complexity.

Table 8. Squared subspace distance m^2 by link function (mean $\pm$ s.e. over n, p , covariance, replications). **Bold**: best; underline: second best.

Link	OPG	RMAVE	OPG-kNN	SMAVE
Polynomial	0.75 $\pm$.06	0.19 $\pm$.03	0.72 $\pm$.06	0.07$\pm$.01
Sinusoidal	0.70 $\pm$.07	0.27 $\pm$.03	0.63 $\pm$.06	0.15$\pm$.02
Exponential	0.71 $\pm$.06	0.22 $\pm$.03	0.66 $\pm$.06	0.09$\pm$.01
Interaction	0.67 $\pm$.07	0.02 $\pm$.01	0.66 $\pm$.07	0.02$\pm$.00
Rational	0.71 $\pm$.06	0.08 $\pm$.02	0.63 $\pm$.06	0.03$\pm$.00

D.2. Results by covariance structure

Table 9 compares performance under identity and AR(1) covariance structures. All methods exhibit similar relative rankings under both settings. SMAVE achieves comparable error under identity (0.08 $\pm$.01) and AR(1) correlation (0.06 $\pm$.01), indicating that k -NN localization is robust to correlation structure without requiring explicit covariance adaptation.

Table 9. Squared subspace distance m^2 by covariance structure (mean $\pm$ s.e. over link functions, (n, p) , replications). **Bold**: best; underline: second best.

Covariance	OPG	RMAVE	OPG-kNN	SMAVE
Identity	0.70 $\pm$.04	0.16 $\pm$.02	0.62 $\pm$.04	0.08$\pm$.01
AR(1)	0.72 $\pm$.04	0.15 $\pm$.02	0.71 $\pm$.04	0.06$\pm$.01

D.3. Convergence curves

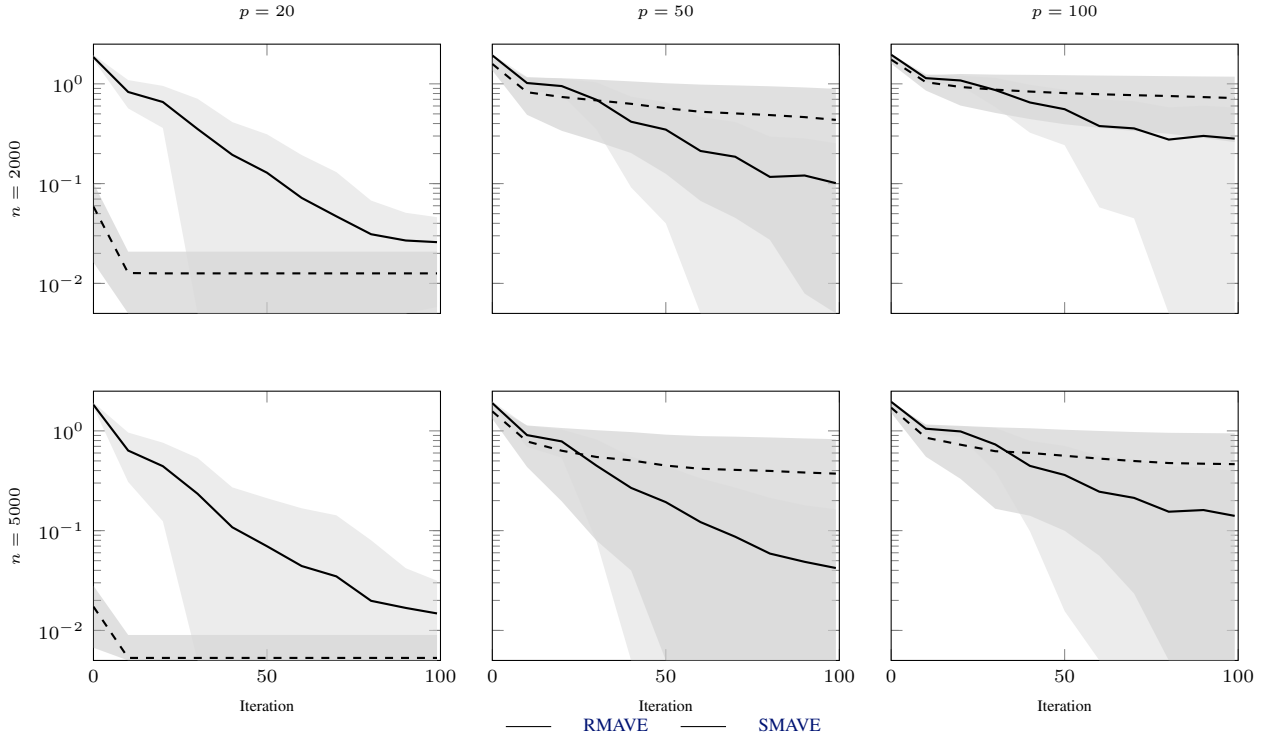


Figure 2. Convergence of subspace error m^2 for $n \in \{2000, 5000\}$ (rows) and $p \in \{20, 50, 100\}$ (columns). Each curve shows the mean ± 1 std aggregated over 5 link functions, 2 covariance structures, and 10 replications (100 runs total). At low dimension ($p = 20$), RMAVE converges within a few iterations; at higher dimensions ($p \geq 50$), SMAVE achieves 2–9 $\times$ lower final error.

E. Real-data experiments: test MSE by candidate dimension

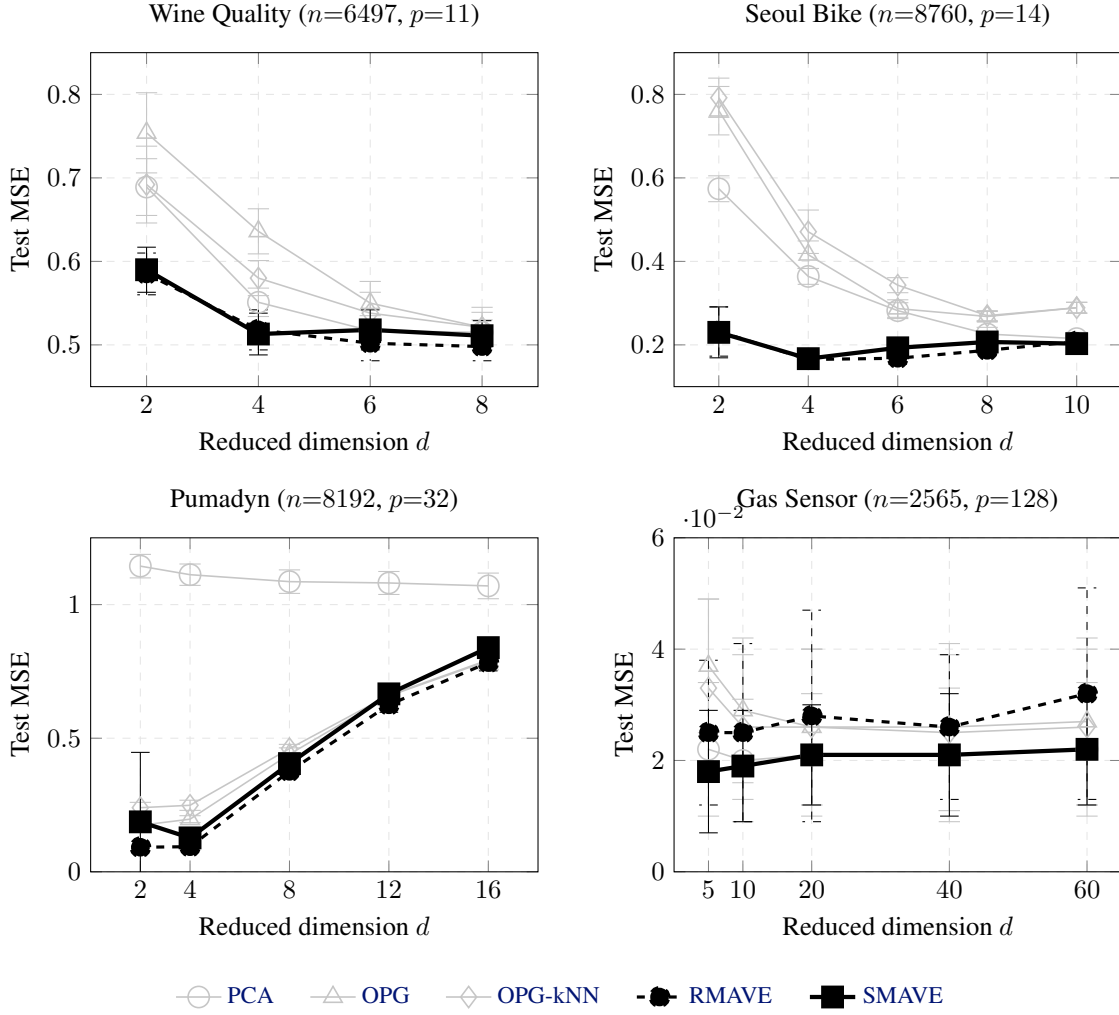


Figure 3. Test MSE vs. reduced dimension d across four datasets. Baselines (PCA, OPG, OPG-kNN) in gray; iterative methods (RMAVE, SMAVE) in black.